

Abstract

An article usually includes an abstract, a concise summary of the work covered at length in the main body of the article.

I. INTRODUCTION

Intro on the subject.

II. METHOD

A. Model

We consider a stochastic generalized Lotka-Volterra model on a sparse graph, describing the dynamics of an interacting community of N species, labeled by $i = 1, \dots, N$. The species interacts on an undirected random graph G with symmetric adjacency matrix $A = \{a_{ij}\}_{i,j=1,\dots,N}$, meaning that species i and j are interacting if $a_{ij} = a_{ji} = 1$, otherwise the matrix elements are zero. The graph have a set of degrees $\{k_i\}_{i=1,\dots,N}$, distributed according to the degree distribution $p_{\text{deg}}(k)$ with average degree K . The time-dependent number density of individuals of species i is denoted by $n_i(t)$, and evolves in time according to the following coupled stochastic differential equations (SDEs)

$$\frac{d}{dt}n_i = \frac{r_i}{K_i}n_i \left[f_i(n_i) - \sum_{j(\neq i)=1}^N a_{ij}J_{ij}n_j \right] + \xi_i + \lambda, \quad (1)$$

where r_i is the intrinsic growth rate of species i , K_i is the carrying capacity, corresponding to the equilibrium abundance to which species i would converge in the absence of interactions. The function $f_i(x)$ describes the growth of species i without interactions with other species. The term λ is a homogenous immigration rate for all species, ensuring the presence of all invadable species. The noise term ξ_i is a Gaussian multiplicative noise with zero mean and variance $\langle \xi_i(t)\xi_{i'}(t') \rangle = 2T\delta_{ii'}g(n_i(t))\delta(t-t')$ where we consider two types of noise sources: demographic noise, with $g(x) = x$, describing the fluctuations in the number of individuals due to birth-death processes, and environmental noise, with $g(x) = x^2$, describing the fluctuations of the environment. The parameter T is the noise intensity, which can be related to the temperature of the system. The SDEs have to be intended in the Itô

sense, since it correctly describes the absorbing state at $n_i = 0$, which is stable under noise fluctuations.

The couplings J_{ij} represent the (per capita) effect of species on one another. A negative coefficient J_{ij} indicates a competitive effect of species j on species i . We consider here gaussian couplings with average $\overline{J_{ij}} = \overline{J_{ji}} = m/K$, variance $\text{Var}(J_{ij}) = \text{Var}(J_{ji}) = \sigma^2/K$ and correlation $\text{Corr}(J_{ij}, J_{ji}) = \gamma$. The overbar $\overline{\cdots}$ represents the average over the random couplings ensemble. The scaling of the moments with the average degree of the graph K is needed to produce a well-defined fully connected limit $K \rightarrow \infty$. The parameter $-1 \leq \gamma \leq 1$, being related to the correlation between the interaction coefficients J_{ij} and J_{ji} , measures the fraction of predator-prey pairs. Such a predator-prey pair consists of two species i and j for which J_{ij} and J_{ji} have opposite signs. If for example J_{ij} is positive and J_{ji} is negative, the presence of species i has a detrimental effect on species j , while the presence of species j is beneficial for species i . For any pair $i < j$ we can write

$$J_{ij} = \frac{m}{K} + \frac{\sigma}{\sqrt{K}} z_{ij}, \quad J_{ji} = \frac{m}{K} + \frac{\sigma}{\sqrt{K}} z_{ji}, \quad (2)$$

where z_{ij} and z_{ji} are Gaussian random variables with $\overline{z_{ij}} = 0$, $\overline{z_{ij}^2} = 1$, $\overline{z_{ij} z_{ji}} = \gamma$. For $\gamma = 1$ we have $z_{ij} = z_{ji}$ with probability one, for $\gamma = 0$ z_{ij} and z_{ji} are uncorrelated, and for $\gamma = -1$ we have $z_{ij} = -z_{ji}$ with probability one. The latter, $\gamma = -1$, corresponds to having all the interactions of the predator-prey type. Increasing the value of γ this fraction decrease, until the opposite case, $\gamma = 1$ of no predator-prey type interactions. The other interactions can be of a mutualistic type (J_{ij} and J_{ji} both positive) or strictly competitive (J_{ij} and J_{ji} both negative).

Without loss of generality, we will focus on the case where r_i and K_i are constant for all species, $r_i = r$ and $K_i = K$. We will also consider the case where $f_i(x) = x(1 - x)$, which is a common choice in the literature. The model can be written in an adimensional form by rescaling the time by rK and introducing the relative abundance $x_i = n_i/K$. Then, by rescaling the immigration rate by rK , we obtain the following set of equations

$$\frac{d}{dt} x_i = x_i \left[1 - x_i - \sum_{j(\neq i)=1}^N a_{ij} J_{ij} x_j \right] + \eta_i + \lambda, \quad (3)$$

where the noise term η_i is a Gaussian multiplicative noise with zero mean and variance $\langle \eta_i(t) \eta_{i'}(t') \rangle = 2T \delta_{ii'} g(x_i(t)) \delta(t - t')$. The noise intensity T needs to be rescaled by $r^2 K$ in the case of intrinsic noise, and by r^2 in the case of extrinsic noise.

B. Equilibrium Hamiltonian

We focus on the case of demographic noise with symmetric couplings, $\gamma = 1$, which admits a description in terms of an equivalent equilibrium Hamiltonian [1]. The equilibrium Hamiltonian is defined as the logarithm of the probability distribution of the system in the stationary state, which can be computed from the Fokker-Planck equation associated to the SDEs

$$\frac{\partial}{\partial t} P_{\text{eq}}(\mathbf{x}) = \sum_{i=1}^N \left\{ -\frac{\partial}{\partial x_i} [\mathcal{D}_i(\mathbf{x}) P_{\text{eq}}(\mathbf{x})] + T \frac{\partial^2}{\partial x_i^2} [g(x_i) P_{\text{eq}}(\mathbf{x})] \right\} = 0, \quad (4)$$

where we have defined the deterministic drift term as

$$\mathcal{D}_i(\mathbf{x}) = x_i \left(1 - x_i - \sum_{j(\neq i)=1}^N a_{ij} J_{ij} x_j \right) + \lambda. \quad (5)$$

The equilibrium distribution must satisfy the detailed balance condition, which in the case of the SDEs reads for every species i

$$\frac{\partial}{\partial x_i} P_{\text{eq}}(\mathbf{x}) = \left[\frac{\mathcal{D}_i(\mathbf{x})}{T} - g'(x_i) \right] \frac{P_{\text{eq}}(\mathbf{x})}{g(x_i)}. \quad (6)$$

The equilibrium distribution can be written as a Boltzmann distribution, i.e. $P_{\text{eq}}(\mathbf{x}) = Z^{-1} \exp(-H/T)$, where Z is the usual partition function and H is the equilibriumlike Hamiltonian, which solves the partial differential equation

$$\frac{\partial}{\partial x_i} H(\mathbf{x}) = -\frac{1}{g(x_i)} [\mathcal{D}_i(\mathbf{x}) - T g'(x_i)]. \quad (7)$$

Hence the Hamiltonian for the demographic noise case reads

$$H(\mathbf{x}) = \sum_{i=1}^N \left[-x_i + \frac{1}{2} x_i^2 + (T - \lambda) \log x_i \right] + \sum_{i < j} a_{ij} J_{ij} x_i x_j. \quad (8)$$

C. Cavity ansatz

The equilibrium Hamiltonian can be used to compute the distribution of species i in the stationary state, $P_i(x_i)$, by marginalizing the equilibrium distribution over all the other species. If the underlying interaction graph is locally tree-like, which is the case for sparse random graphs, the distribution $P_i(x_i)$ can be computed using the cavity method, by noting the factorization of the equilibrium distribution as

$$P_{\text{eq}}(\mathbf{x}) = \prod_{i=1}^N e^{-H_i(x_i)/T} \prod_{(i,j) \in E} e^{-H_{ij}(x_i, x_j)/T}, \quad (9)$$

$$H_i(x_i) = -x_i + \frac{1}{2}x_i^2 + (T - \lambda) \log x_i, \quad (10)$$

$$H_{ij}(x_i, x_j) = J_{ij}x_ix_j. \quad (11)$$

83 In order to avoid numerical instabilities when the temperature T is small, we rescale the
84 abundances x_i by the temperature, $\tilde{x}_i = x_i/T$. The rescaled equilibrium distribution reads

$$\tilde{P}_{\text{eq}}(\tilde{\mathbf{x}}) = T^{N\frac{\lambda}{T}} \prod_{i=1}^N e^{-\tilde{H}_i(\tilde{x}_i)} \prod_{(i,j) \in E} e^{-\tilde{H}_{ij}(\tilde{x}_i, \tilde{x}_j)}, \quad (12)$$

85 where we have defined the rescaled local and interacting Hamiltonians as

$$\tilde{H}_i(\tilde{x}_i) = -\tilde{x}_i + \frac{T}{2}\tilde{x}_i^2 + \left(1 - \frac{\lambda}{T}\right) \log \tilde{x}_i, \quad (13)$$

$$\tilde{H}_{ij}(\tilde{x}_i, \tilde{x}_j) = T J_{ij} \tilde{x}_i \tilde{x}_j. \quad (14)$$

86 For the sake of simplicity, we will drop the tilde notation in what follows unless specified
87 otherwise.

88 Due to the factorization of the equilibrium distribution, we can put forward the following
89 cavity ansatz

$$c_{i \setminus j}(x_i) = \frac{1}{Z_{i \setminus j}} e^{-H_i(x_i)} \prod_{k \in \partial i \setminus j} \int dx_k c_{k \setminus i}(x_k) e^{-H_{ki}(x_k, x_i)}, \quad (15)$$

90 where $Z_{i \setminus j}$ is the normalization constant, and $\partial i \setminus j$ is the set of neighbors of species i
91 excluding species j . The cavity distribution $c_{i \setminus j}(x_i)$ is the distribution of species i when
92 species j is removed from the graph.

93 The marginal distribution $P_i(x_i)$ can be approximated within the cavity ansatz as

$$c_i(x_i) = \frac{1}{Z_i} e^{-H_i(x_i)} \prod_{k \in \partial i} \int dx_k c_{k \setminus i}(x_k) e^{-H_{ki}(x_k, x_i)}, \quad (16)$$

94 where Z_i is the normalization constant, and ∂i is the set of neighbors of species i .

95 D. Small coupling / large connectivity / large noise expansion

96 The cavity method, while being exact on locally tree-like graphs, is not directly applicable
97 since it involves the computation of the integrals over the cavity distributions. We can

98 however expand the cavity distributions in powers of the couplings, and truncate the
 99 expansion at the second order

$$c_{i \setminus j}(x_i) \approx \frac{e^{-H_i(x_i)}}{Z_{i \setminus j}} \prod_{k \in \partial i \setminus j} \int dx_k c_{k \setminus i}(x_k) \left(1 - H_{ki} + \frac{1}{2} H_{ki}^2\right). \quad (17)$$

100 Introducing the average over the cavity distributions as

$$\langle \dots \rangle_{i \setminus j} = \int dx_i c_{i \setminus j}(x_i) \dots, \quad (18)$$

101 we can write the expanded cavity distribution as

$$c_{i \setminus j}(x_i) \approx \frac{e^{-H_i(x_i)}}{Z_{i \setminus j}} \prod_{k \in \partial i \setminus j} \left(1 - J_{ki} T \mu_{k \setminus i} x_i + \frac{1}{2} J_{ki}^2 T^2 \mu_{k \setminus i}^{(2)} x_i^2\right), \quad (19)$$

102 where we have defined the cavity mean abundances $\mu_{i \setminus j} = \langle x_i \rangle_{i \setminus j}$ and the cavity second
 103 moment $\mu_{i \setminus j}^{(2)} = \langle x_i^2 \rangle_{i \setminus j}$.

104 By re-exponentiating the terms in the product, we can write the cavity distribution as

$$c_{i \setminus j}(x_i) \approx \frac{1}{Z_{i \setminus j}} \exp \left(-H_i(x_i) - \sum_{k \in \partial i \setminus j} J_{ki} T \mu_{k \setminus i} x_i + \frac{1}{2} \sum_{k \in \partial i \setminus j} J_{ki}^2 T^2 q_{k \setminus i} x_i^2 \right), \quad (20)$$

105 where we have defined the cavity variance $q_{i \setminus j} = \mu_{i \setminus j}^{(2)} - \mu_{i \setminus j}^2$. Inserting the definition of the
 106 local hamiltonian Eq. (10), we can write the cavity distribution as

$$\begin{aligned} c_{i \setminus j}(x_i) \approx \frac{1}{Z_{i \setminus j}} \exp & \left[\left(1 - \sum_{k \in \partial i \setminus j} J_{ki} T \mu_{k \setminus i}\right) x_i - \frac{1}{2} \left(T - \sum_{k \in \partial i \setminus j} J_{ki}^2 T^2 q_{k \setminus i}\right) x_i^2 \right. \\ & \left. + \left(\frac{\lambda}{T} - 1\right) \log x_i \right]. \end{aligned} \quad (21)$$

107 The cavity measure gets approximated by an effective non interacting equilibrium measure
 108 with effective Hamiltonian

$$H_{i \setminus j}^{\text{eff}}(x_i) = -a_{i \setminus j} x_i + \frac{1}{2} b_{i \setminus j} x_i^2 + (\rho - 1) \log x_i, \quad (22)$$

109 with the normalization constant $Z_{i \setminus j}$ approximated by

$$\begin{aligned} Z_{i \setminus j}^{\text{eff}} = \sqrt{\frac{2^{\rho-2}}{b_{i \setminus j}^{\rho+1}}} & \left[\sqrt{b_{i \setminus j}} \Gamma\left(\frac{\rho}{2}\right) {}_1F_1\left(\frac{\rho}{2}, \frac{1}{2}, \frac{a_{i \setminus j}^2}{2b_{i \setminus j}}\right) \right. \\ & \left. + \sqrt{2} a_{i \setminus j} \Gamma\left(\frac{1+\rho}{2}\right) {}_1F_1\left(\frac{1+\rho}{2}, \frac{3}{2}, \frac{a_{i \setminus j}^2}{2b_{i \setminus j}}\right) \right], \end{aligned} \quad (23)$$

where we have defined the shorthand notation $a_{i \setminus j} = 1 - \sum_{k \in \partial i \setminus j} J_{ki} T \mu_{k \setminus i}$, $b_{i \setminus j} = T - \sum_{k \in \partial i \setminus j} J_{ki}^2 T^2 q_{k \setminus i}$, and $\rho = \lambda/T$. The function $\Gamma(z)$ is the gamma function, and ${}_1F_1(a, b, z)$ is the Kummer's confluent hypergeometric function.

We can therefore obtain a closed set of equations for the cavity mean abundances and variances by requiring the cavity distribution to be a fixed point of the cavity equations. The cavity equations read

$$\mu_{i \setminus j} = \frac{1}{\sqrt{b_{i \setminus j}}} \frac{a_{i \setminus j} \rho \Gamma\left(\frac{\rho}{2}\right) {}_1F_1\left(1 + \frac{\rho}{2}, \frac{3}{2}, \frac{a_{i \setminus j}^2}{2b_{i \setminus j}}\right) + \sqrt{2b_{i \setminus j}} \Gamma\left(\frac{1+\rho}{2}\right) {}_1F_1\left(\frac{1+\rho}{2}, \frac{1}{2}, \frac{a_{i \setminus j}^2}{2b_{i \setminus j}}\right)}{\sqrt{b_{i \setminus j}} \Gamma\left(\frac{\rho}{2}\right) {}_1F_1\left(\frac{\rho}{2}, \frac{1}{2}, \frac{a_{i \setminus j}^2}{2b_{i \setminus j}}\right) + \sqrt{2} a_{i \setminus j} \Gamma\left(\frac{1+\rho}{2}\right) {}_1F_1\left(\frac{1+\rho}{2}, \frac{3}{2}, \frac{a_{i \setminus j}^2}{2b_{i \setminus j}}\right)}, \quad (24)$$

$$\mu_{i \setminus j}^{(2)} = \frac{1}{b_{i \setminus j}} \frac{\sqrt{b_{i \setminus j}} \rho \Gamma\left(\frac{\rho}{2}\right) {}_1F_1\left(1 + \frac{\rho}{2}, \frac{1}{2}, \frac{a_{i \setminus j}^2}{2b_{i \setminus j}}\right) + 2^{3/2} a_{i \setminus j} \Gamma\left(\frac{3+\rho}{2}\right) {}_1F_1\left(\frac{3+\rho}{2}, \frac{3}{2}, \frac{a_{i \setminus j}^2}{2b_{i \setminus j}}\right)}{\sqrt{b_{i \setminus j}} \Gamma\left(\frac{\rho}{2}\right) {}_1F_1\left(\frac{\rho}{2}, \frac{1}{2}, \frac{a_{i \setminus j}^2}{2b_{i \setminus j}}\right) + \sqrt{2} a_{i \setminus j} \Gamma\left(\frac{1+\rho}{2}\right) {}_1F_1\left(\frac{1+\rho}{2}, \frac{3}{2}, \frac{a_{i \setminus j}^2}{2b_{i \setminus j}}\right)}. \quad (25)$$

We can now implement those equations in a message passing algorithm, which allows to compute the cavity mean abundances and variances iteratively. The algorithm is initialized with random values for the cavity mean abundances and variances, and iterated until convergence. The fixed point of the cavity equations gives the cavity mean abundances and variances, which can be used to compute the average abundance of species i in the stationary state.

When dealing with disordered interactions, one can make use of a population dynamics algorithm to compute the histogram of the cavity mean abundances and variances. The population dynamics algorithm is initialized with a population of N_{pop} samples of the cavity mean abundances and variances, and iterated for a number of generations until convergence.

E. Replica symmetric ansatz

When the system is in the replica symmetric (RS) phase, the cavity distributions become independent of the species indices. We can introduce the RS cavity distribution of species that have a particular degree k in the original community as $c_k^{\text{cav}}(x)$. This corresponds to the equilibrium distribution of abundances x of a typical species with degree k in the cavity graph. Performing the average over the disorder of Eq. (15), we obtain

$$c_k^{\text{cav}}(x) \propto e^{-H_{\text{loc}}(x)} \left[\sum_{k'} p_{\text{cav}}(k') \int dy c_{k'}^{\text{cav}}(y) e^{-H_{\text{int}}(x,y)} \right]^k, \quad (26)$$

where $p_{\text{cav}}(k) = (k+1)p(k+1)/K$ is the cavity degree distribution, sometimes called excess degree distribution, and the overline $\overline{\cdots}$ represents the average over the disordered couplings J . The local Hamiltonian $H_{\text{loc}}(x)$ and the interacting Hamiltonian $H_{\text{int}}(x, y)$ are defined as

$$H_{\text{loc}}(x) = -x + \frac{T}{2}x^2 + \left(1 - \frac{\lambda}{T}\right) \log x, \quad (27)$$

$$H_{\text{int}}(x, y) = TJxy. \quad (28)$$

The average over the disorder of the interacting Hamiltonian is easily computed, being it a Gaussian integral

$$\overline{e^{-H_{\text{int}}(x, y)}} = \int dJ e^{-TJxy} \frac{1}{\sqrt{2\pi\sigma^2/K}} e^{-\frac{1}{2} \frac{(J-m/K)^2}{\sigma^2/K}} = e^{-\frac{1}{K} (mTxy - \frac{\sigma^2}{2} T^2 x^2 y^2)}. \quad (29)$$

Inserting the result in Eq. (26), and expanding the exponential at second order in the coupling strength, we obtain [MT: Check dimensionality!!!!]

$$c_k^{\text{cav}}(x) \approx e^{-H_{\text{loc}}(x)} \left\{ \sum_{k'} p_{\text{cav}}(k') \left[1 - \frac{m}{K} T \mu_{k'}^{\text{cav}} x + \frac{\sigma^2}{2K} T^2 \mu_{k'}^{\text{cav},(2)} x^2 \right] \right\}^k, \quad (30)$$

where we have defined the cavity mean abundances and second moments as $\mu_k^{\text{cav}} = \langle x \rangle_k^{\text{cav}}$ and $\mu_k^{\text{cav},(2)} = \langle x^2 \rangle_k^{\text{cav}}$, with the average over the disorder averaged cavity distribution c_k^{cav} denoted by $\langle \dots \rangle_k^{\text{cav}}$. Re-exponentiating the terms in the product, we obtain

$$c_k^{\text{cav}}(x) \approx e^{-H_{\text{loc}}(x)} \left[\sum_{k'} p_{\text{cav}}(k') e^{-\frac{1}{K} (mT \mu_{k'}^{\text{cav}} x - \frac{\sigma^2}{2} T^2 \mu_{k'}^{\text{cav},(2)} x^2)} \right]^k. \quad (31)$$

Let's focus on the case of a Random Regular Graph (RRG) with degree K , which is a particular case of a sparse graph. The degree distribution of a RRG is a Kronecker delta function, $p_{\text{deg}}(k) = \delta_{k,K}$. The RS cavity equations simplify to

$$c^{\text{cav}}(x) \approx e^{-H_{\text{loc}}(x)} e^{-\frac{K-1}{K} (mT \mu^{\text{cav}} x - \frac{\sigma^2}{2} T^2 \mu^{\text{cav},(2)} x^2)}, \quad (32)$$

where we have defined $c_{K-1}^{\text{cav}} = c^{\text{cav}}$, and the cavity mean abundances and second moments as $\mu_{K-1}^{\text{cav}} = \mu^{\text{cav}}$ and $\mu_{K-1}^{\text{cav},(2)} = \mu^{\text{cav},(2)}$ for shortness of notation.

In the fully connected limit, the cavity equations simplify to their mean-field form, which can be obtained by taking the limit $K \rightarrow \infty$ in the cavity equation Eq. (32)

$$c_{\text{MF}}(x) \approx \frac{1}{Z_{\text{MF}}} e^{-H_{\text{loc}}(x)} e^{-\left(mT \mu_{\text{MF}} x - \frac{\sigma^2}{2} T^2 \mu_{\text{MF}}^{(2)} x^2\right)} = \frac{1}{Z_{\text{MF}}} e^{-H_{\text{MF}}(x)}, \quad (33)$$

where we have defined the mean-field effective Hamiltonian $H_{\text{MF}}(x)$ as

$$H_{\text{MF}}(x) = -(1 - mT \mu_{\text{MF}}) x + \frac{1}{2} \left(T - \sigma^2 T^2 \mu_{\text{MF}}^{(2)} \right) x^2 + \left(1 - \frac{\lambda}{T} \right) \log x. \quad (34)$$

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